

Attention-Enhanced Pix2Pix GAN for Real-Time Smoke Removal

Progress Report: TiHAN Project on Efficient Road Object Detection in Extreme Weather Conditions

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Abstract

Autonomous Driving Assistance Systems (ADAS) faces critical challenges in extreme weather conditions like fog, smoke, rain, snow, or storms. This report presents the *Attention enhanced Pix2Pix GAN*, that integrates Attention Gates into the skip connections between the generator and discriminator GAN structure. Attention induces selective filtering of features along with preserving the scene structure, which has proved an improvement over the previous proposed Pix2Pix GAN.

1 METHODOLOGY

1.1 Architecture Overview

We adopt the Pix2Pix conditional GAN (cGAN) framework, using a UNet generator. Standard skip connections pass all low-level features to the decoder. To prevent which, we replace standard skips with **Attention Gates (AGs)** that suppress irrelevant features transfer.

1.2 Attention Mechanism

Working:

- The encoder provides features x (K, V) that can be useful or noisy.
- The decoder provides a gating signal g (Q) that indicates what kind of information it needs.
- With these input, Attention gate produces a mask, α between 0 and 1

Mask keeps important features and discards irrelevant ones. Filtered output can be represented as:

$$x_{\text{Att}} = \alpha \cdot x$$

Here,

- $\alpha \approx 1$ means “retain this feature” (e.g., vehicles, road edges).
- $\alpha \approx 0$ means “detain this feature” (e.g., smoke regions).

Thus, Attention Gates prevent smoke-related artifacts from being passed through skip connections and help the generator focus on relevant scene structures.

1.3 Training Objective

The total loss combines the adversarial objective with L1 reconstruction:

$$L_{\text{Total}} = L_{\text{GAN}}(G, D) + \lambda L_{\text{L1}}(G), \quad (1)$$

We tend to achieve two things:

1. Minimize L_{Total}
2. High structural fidelity: preserving actual shapes, edges and layout Lambda is set 100 to penalize generator for distortions in generated images, pushing generator towards accurate reconstructions.

2 IMPLEMENTATION DETAILS

2.1 Dataset

We used the smoke dataset made available at [this Google Drive folder](#), from [Kaggle](#)

2.2 Training Setup

The [model](#) was implemented in PyTorch using:

- **Epochs:** 100
- **Optimizer:** Adam ($\beta_1 = 0.5$, $\beta_2 = 0.999$)
- **Learning Rate:** 1×10^{-4}
- **Hardware:** GPU (Target deployment: NVIDIA Jetson / Edge TPU)

3 RESULTS

3.1 Qualitative Analysis



Figure 1: Input smoke-degraded frame (left) and restored output (right) using the proposed Attention-Pix2Pix model.

3.2 Quantitative Metrics

Table 1: Performance comparison on test set

Method	PSNR (dB) $\uparrow$	SSIM $\uparrow$
Pix2Pix (Vanilla)	21.45	0.72
Ours (Attention)	25.30	0.79

4 CONCLUSION AND FUTURE WORK

We successfully implemented an attention-enhanced Pix2Pix model for smoke removal in ADAS vision pipelines. The model significantly improves both perceptual quality and structural restoration.

Future directions include:

1. **Versatile and efficient model:** Primary aim from here, is to adapt model to other weather conditions viz., snow, rain, fog and sand storms. And, implement an efficient architecture at layer-level towards building a robust model for both deweathering and object detection.
2. **INT8 Quantization:** For real-time deployment on edge hardware.
3. **Robustness Analysis:** Extending evaluation under adversarial or extreme-variance inputs.

References

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